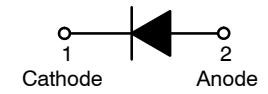
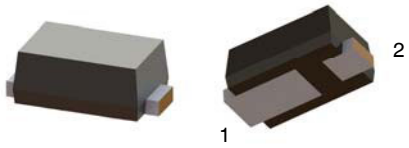
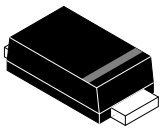
Surface Mount Rectifiers, 1 A, 400 V - 600 V

S1GHE, S1JHE



Rectifier



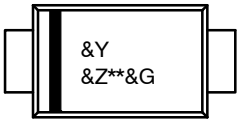


SOD−323EP
CASE 477AD

Features

- Low Profile Package with < 0.75 mm Package Height
- High Efficiency
- Moisture Sensitivity Level 1 per J−STD−020
- Glass Passivated Chip Junction
- UL Flammability 94V−0 Classification
- Green Mold Compound
- NRV Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC−Q101 Qualified and PPAP Capable
- These Devices are Pb−Free and are RoHS Compliant

MARKING DIAGRAM



Band Indicates Cathode

&Y	= Binary Calendar Year Coding Scheme
&Z	= Assembly Plant Code
**	= Specific Device Code − (A5, A7)
&G	= Single Digit Weekly Data Code

Specifications

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)

Symbol	Parameter	Value		Unit
		S1GHE	S1JHE	
V_{RRM}	Maximum Repetitive Peak Reverse Voltage	400	600	V
$I_{F(AV)}$	Maximum Average Forward Rectified Current	1		A
I_{FSM}	Peak Forward Surge Current, 8.3 ms Single Half Sine−Wave Superimposed on Rated Load	20		A
T_J	Operating Junction Temperature Range	−55 to +175		°C
T_{STG}	Storage Temperature Range	−55 to +175		°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

ORDERING INFORMATION

Part Number	Device Code Marking	Package	Shipping†
S1GHE	A5	SOD−323EP (Pb−Free/Halogen Free)	3000 / Tape & Reel
NRVS1GHE			
S1JHE	A7	SOD−323EP (Pb−Free/Halogen Free)	3000 / Tape & Reel
NRVS1JHE			

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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THERMAL CHARACTERISTICS (T_A = 25°C unless otherwise noted) (Note 1)

Symbol	Characteristic	Value	Unit
Ψ_{JL}	Junction to Lead Thermal Resistance Thermocouple Soldered to Cathode	26.5	°C/W
$R_{\theta JA}$	Junction to Ambient Thermal Resistance	200	°C/W

1. Per JESD51−3 Recommended Thermal Test Board. Device mounted on FR−4 PCB, board size = 76.2 mm x 114.3 mm.

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_F	Instantaneous Forward Voltage (Note 2)	I_F = 1 A		0.96	1.1	V
I_R	Reverse Current at Rated V_R	T_J = 25°C		0.02	1	μA
		T_J = 125°C		10.35	50	
T_{rr}	Reverse Recovery Time	I_F = 0.5 A, I_R = 1.0 A, I_{rr} = 0.25 A		782		ns
C_J	Junction Capacitance	V_R = 4.0 V, f = 1 MHz		3		pF

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

2. Pulse test with PW = 300 μs, 1% duty cycle.

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TYPICAL PERFORMANCE CHARACTERISTICS

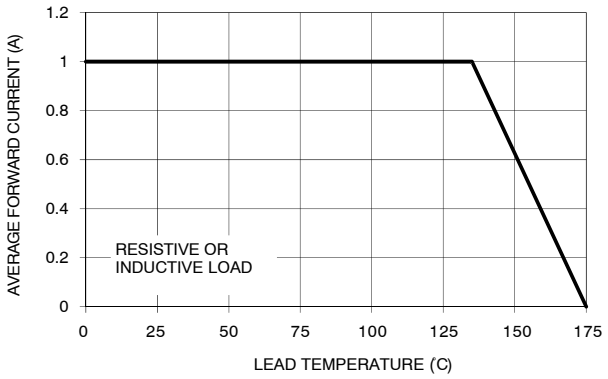


Figure 1. Forward Current Derating Curve



Figure 2. Maximum Non−Repetitive Forward Surge Current

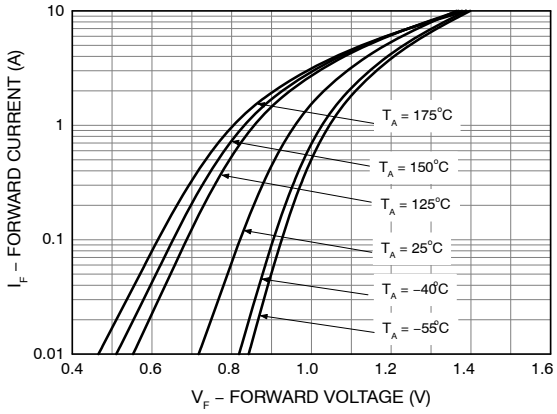


Figure 3. Typical Forward Characteristics

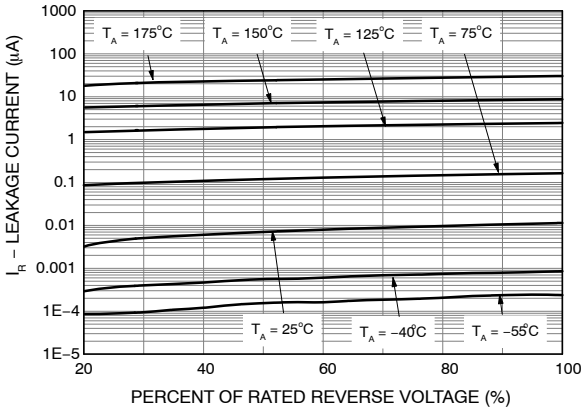


Figure 4. Typical Reverse Characteristics

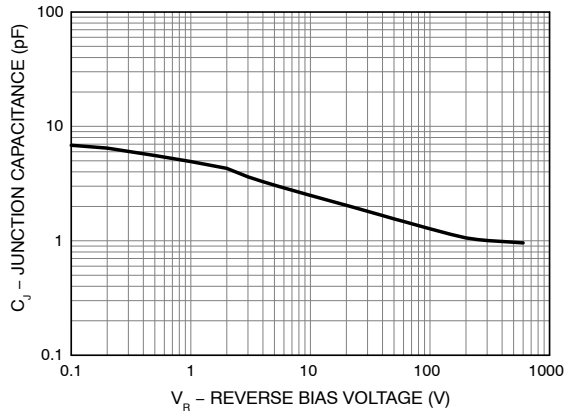


Figure 5. Typical junction Capacitance

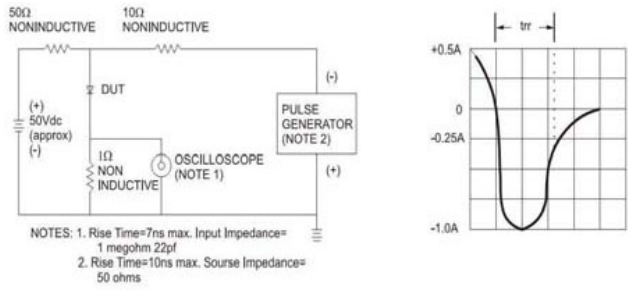
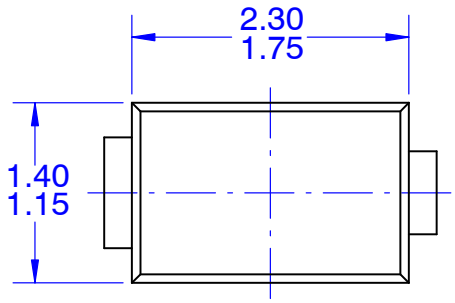


Figure 6. Reverse Recovery Time Characteristics and Test Circuit Diagram

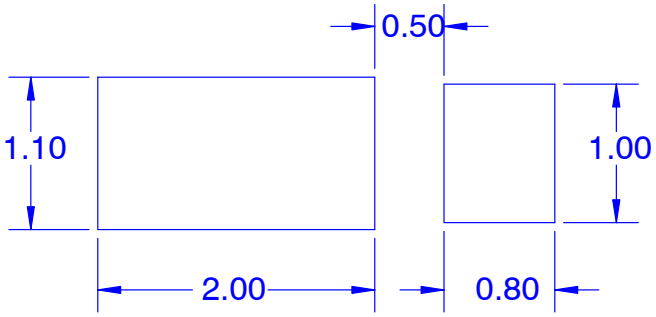
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SOD−323EP
CASE 477AD
ISSUE O

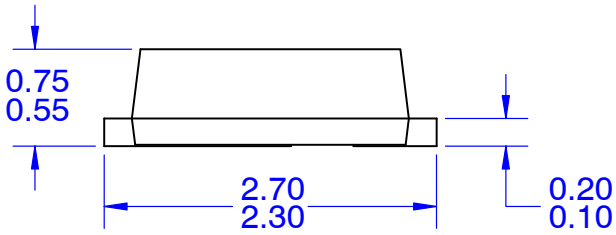
DATE 31 AUG 2016



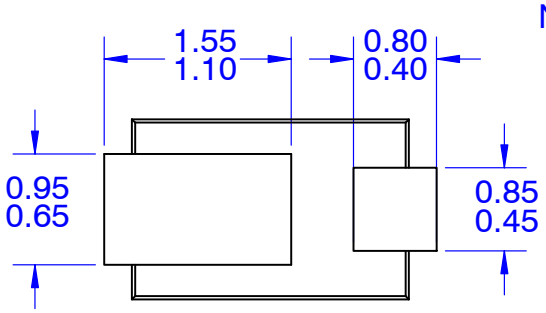
TOP VIEW



LAND PATTERN RECOMMENDATION



FRONT VIEW



BOTTOM VIEW

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